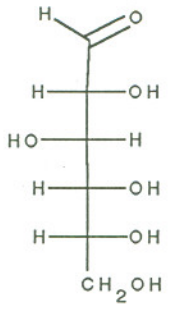
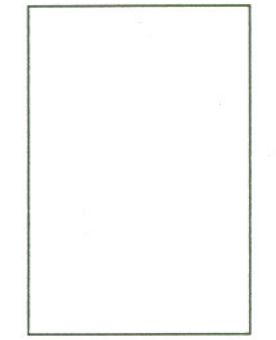
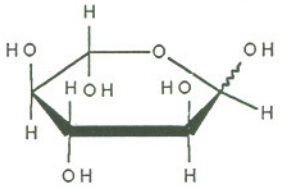
ORGANIC CHEMISTRY CUME

DEC 1ST 1990

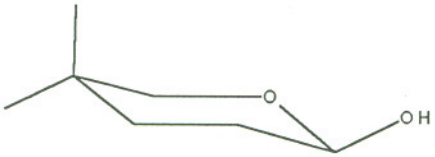
1. Sugars can be drawn in a number of different manners to help illustrate different aspects of their chemistry. Shown below is the extended form of D-(+)-glucose and the Haworth formula representation of L-(-)-glucose. Fill in the boxes by drawing the extended form of L-(-)-glucose and the Haworth formula representation of D-(+)-glucose. (10pts)

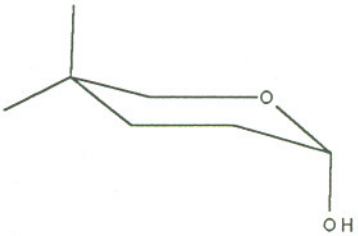




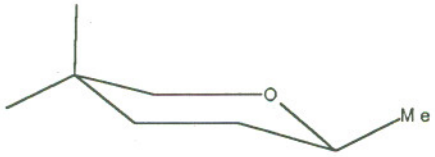


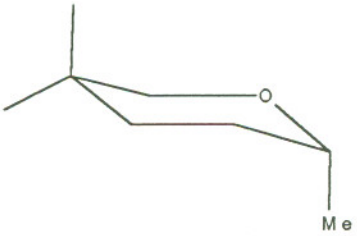
2. Shown below are the two hemiacetal forms of a simple hydroxyaldehyde. The hemiacetal OH unit may be either axial or equatorial with respect to the tetrahydropyran ring system. Which of the two conformations would you expect to be the most stable (lowest in energy)? Justify your answer with respect to the 3-dimensional structure of these species. (20pts)





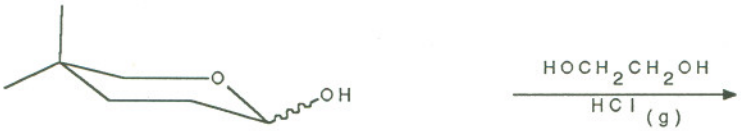
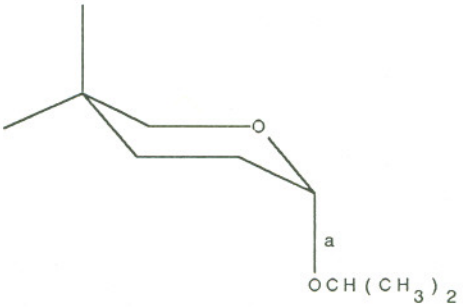
3. Illustrated below are two cyclic ether structures that are analogous to the two structures that you analyzed above. Which of these two conformations would you expect to have the lowest energy? Justify your answer with reference to your previous answer. (10pts)





4. Draw the mechanism of the following transformation, clearly defining each stage of the mechanism. Show whether each individual step is reversible or irreversible and use arrows to indicate the motion of the electrons concerned in the reaction. [NO credit will be given if arrows are not utilized] (25pts)

5. What is the product of the following reaction? (10pts)



6. Draw a graph showing the potential energy changes that accompany a 360° rotation of the C-O bond labelled a. No numerical values for the energy are required, however you should clearly indicate the position of the ethyl unit at appropriate points in the rotation. (25pts)